

Steam Deck OLED review: It's just better

Engadget · Technology · 2023-11-09

<https://www.engadget.com/steam-deck-oled-review-its-just-better-180038030.html?src=rss>

The first thing to know about the Steam Deck OLED is that it's not the Steam Deck 2. The Steam Deck OLED is a mid-cycle refresh from Valve, similar to the Switch OLED upgrade, but there's a lot more going on internally here. Its screen is better, its battery life is better, its thumbsticks are better, the haptics are better, and its chip and thermals are better. One thing that hasn't changed is its form factor — this is still one of the biggest babies on the handheld block.

The new Steam Deck's main upgrade is its OLED screen, which replaces the original LCD and, crucially, has native support for HDR and faster frame rates. The OLED also has smaller bezels, making the display larger at 7.4 inches, compared with the original 7-inch screen.

Valve Valve Steam Deck OLED SCORE 89 Engadget 89 Critics - Not yet scored N/A Users
- Not yet scored N/A Pros Improved battery life

Bright and smooth visuals

Better thermals

Feels less top-heavy

Long charging cable Cons Still ridiculously large

Seriously it's so big

Valve Valve Steam Deck OLED This isn't the Steam Deck 2 some are waiting for, it's just a welcome improvement on an already good device. \$549 at Steam

OLED screen

The HDR OLED display looks fantastic. It features a wider P3 color gamut, pure blacks and a peak brightness of 1,000 nits, compared with the original max of 400 nits. In action, this means the screen's bezels meld with the black of any game, creating a more polished frame for retro aspect ratios and providing room for higher-contrast colors. The screen on the new model doesn't support variable refresh rates (VRR), but it now tops out at 90Hz, while the original could only hit 60Hz. The updated display is smooth, crisp and bright, like a perfect lemon tart. (Even then, I do not recommend licking the screen.)

Battery life

One of the downsides of the first Steam Deck was its short battery life, which maxed out at eight hours, and that was only under ideal conditions. Valve says the new Steam Deck battery will last 30 to 50 percent longer, depending on how you're playing, and it charges faster — the company claims it will go from 20 percent to 80 percent in 45 minutes.

Custom AMD APU

The Steam Deck OLED has a die-shrunk version of the custom AMD APU found in the original Steam Deck — same CPU and GPU, just more efficient. This is the process Sony takes advantage of to build the PS4 Slim and PS5 Slim, but instead of collapsing the Steam Deck around smaller hardware, Valve chose to add a bigger battery and larger fan to its existing chassis. For developers, this means the device has the same performance targets as the original and there's no need to create games specifically for the new hardware. In the end, the device weighs about 30 grams less than the original model, or the equivalent of five quarters.

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In use

Which leads us to the bad news for my tiny hand gang. The Steam Deck OLED is the same monstrous size as the original: 11.7 inches long, 4.5 inches tall and nearly 2 inches thick. The thumbsticks on the new model are actually 1mm taller than the first version. This was one of my main complaints about the Steam Deck when I first reviewed it: I have smaller-than-average hands, and the Steam Deck looks and feels absolutely ridiculous when I'm holding it. It's just massive, no matter how long my manicure gets. However, the OLED model feels more balanced than the original. It's less top-heavy, and it does technically weigh less. These small adjustments have made a noticeable difference during my playtime, and the device feels slightly more manageable as a handheld — or maybe I'm just too distracted by all of its bright, pretty colors to care about the cramping in my palms.

The thumbsticks on the new handheld have an improved texture that shouldn't turn gray after a few months' use, and they feel perfectly serviceable. In my hands-on time so far, I've found that the OLED touchscreen responds immediately to input, and the haptic feedback on the built-in trackpads feels even more precise, populating in reactive pin-pricks under the textured plastic.

I spent a few cozy nights on my couch swapping between the LCD and OLED Steam Decks, and honestly, after I'm done with this review, I'll probably retire my original model. The Steam Deck OLED is sharper and more responsive; indie games, AAA games and plain old Steam menus look so much better on the new hardware.

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Hades is my most-played game on the original Steam Deck, so I used that to test out Valve's battery life claims. Though it doesn't support HDR, a game like Hades pops on the OLED, and its animations look more fluid than on the LCD version. The OLED lasted four hours and 23 minutes before dying. With the same settings (800p with a 60 fps cap and similar brightness) the original model died after two hours and 42 minutes. That's a 62 percent improvement, surpassing Valve's own estimates, though batteries degrade and I've had my LCD Steam Deck since the device's launch in early 2022. In a stress test that involved playing Elden Ring with everything maxed out, the OLED battery lasted two hours

and 20 minutes, about 40 percent longer than the original.

Playing the Steam Decks back-to-back highlighted another significant quality-of-life improvement: heat management. My palms have gotten toasty playing the OLED model, but they haven't broken a sweat like they often do with the original. The new fan may be larger, but it's not any louder, and it clearly works a treat.

The Steam Deck OLED has a new Wi-Fi 6E module and improved antennas that should enable faster downloads for anyone with a compatible router. I don't have a 6E router or multi-gigabit Wi-Fi, but with a regular Wi-Fi 6 router, the top download speed I saw on the new Steam Deck was 562mbps, which was about 10 percent faster than the old Steam Deck, and 10 percent slower than the speeds I get on my MacBook Pro. The device uses Bluetooth 5.3 and adds a dedicated antenna, which should improve functionality when using Wi-Fi and Bluetooth at the same time.

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Even with improvements to the battery, thermals and weight distribution, the Steam Deck OLED is still more of a Wii U than a Vita. It's a beast of a handheld, and it makes the most sense as a portable console to use around the house, rather than a truly travel-ready device. The new model even comes with an extra-long, 2.5m charging cable, designed specifically with couch-based players in mind. I didn't test out the docking station with the new Steam Deck, but Valve recently released a SteamOS update that lets players take advantage of HDR and VRR on external displays, which only drives home the Wii U-ness of it all.

As far as software, you know the drill by now. Valve has rolled out labels for Steam games signifying how well they'll work on the handheld, with four levels: unknown, unsupported, playable and verified. Playable games generally come with a small caveat, like having to deploy the on-screen keyboard, while verified games have been fully optimized for portable play. Currently, more than 10,000 titles on Steam fall under the playable or verified labels, with more added every day.

When will the Steam Deck OLED be available?

The Steam Deck OLED will go on sale on November 16th at 1pm ET, and Valve has units ready to ship that same day. The company has lowered the price of its LCD lineup and will sell through its backstock of 64GB and 512GB models until they're all gone. It'll continue selling the 256GB LCD model at a reduced price of \$400, while the 512GB OLED model will cost \$550 and the 1TB OLED model will cost \$650. There's also a limited edition 1TB OLED version priced at \$680 that comes with a special carrying case, a translucent gray body and orange accents.

Valve

Wrap-up

According to the developers I spoke with at Valve, this is the definitive version of the Steam Deck. The HDR OLED model is what they would have released the first time around, if the technology had existed before 2022. The handheld market has evolved quickly — in just over a year, the Steam Deck proved there's widespread demand for handheld PCs, and its success helped drive chip makers and display manufacturers to build hardware specifically for high-fidelity mobile play. This is great news for Valve and the Steam Deck, and also the other handhelds coming out of companies like Ayaneo, ASUS, Ayn, Logitech, Retroid and Razer. Hell, maybe even Playdate will get an HDR OLED display one day. (Just kidding; Playdate is literally perfect).

So, no, the HDR OLED refresh isn't the Steam Deck 2, but it is a welcome improvement on an already good device. The Steam Deck OLED is the facetuned, photoshopped, spit-shined version of Valve's handheld, and it features tangible, clever improvements. The updated display is the highlight of the device, while things like faster charging, improved antennas and smoother haptics are welcome bonuses. More importantly, this is Valve renewing its commitment to portable PC gaming, and that's a relief to see. The Steam Deck is the granddaddy of handheld PCs and Valve has the resources to continue innovating in this space. For now, the OLED model is a half-step toward the Steam Deck 2, which may be the final version of the device — unless Valve finally learns how to count to three.